

UNOLIFT, DUOLIFT

Installation and operating instructions



English (GB) Installation and operating instructions

Original installation and operating instructions

These installation and operating instructions describe Grundfos Unolift and Duolift lifting stations. They apply to the tank and installation kits. For detailed information on installing and operating the pumps and controller, see the installation and operating instructions supplied with these products.

Sections 1-4 give the information necessary to be able to unpack, install and start up the product in a safe way.

Sections 5-12 give important information about the product, as well as information on service, fault finding and disposal of the product.

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Read this document before installing the product. Installation and operation must comply with local regulations and accepted codes of good practice.

1. General information

1.1 Symbols used in this document

DANGER



Indicates a hazardous situation which, if not avoided, will result in death or serious personal injury.

WARNING



Indicates a hazardous situation which, if not avoided, could result in death or serious personal injury.

CAUTION



Indicates a hazardous situation which, if not avoided, could result in minor or moderate personal injury.

1.2 Symbols on the product



European conformity

2. Receiving the product

2.1 Transporting the product

WARNING



Falling objects

Death or serious personal injury

- Secure the product during transportation to prevent it from tilting or falling down.

CAUTION



Back injury

Minor or moderate personal injury

- Get assistance from another person or use lifting equipment to carry the product.

CAUTION



Crushing of feet

Minor or moderate personal injury

- Wear safety shoes when moving the product.

2.2 Inspecting the product

1. Check that the product is as ordered.
2. Check that no visible parts have been damaged.
3. If parts are damaged or missing, contact your local Grundfos sales company and refer to the product numbers in section [2.3 Scope of delivery](#).

2.3 Scope of delivery

The scope of delivery depends on your order. Pumps, tank and pipes are supplied separately and must be assembled.

Caution

If used for a liquid which is injurious to health or toxic, the pump will be classified as contaminated.

Before the pump is returned to Grundfos for service, the safety declaration at the end of these instructions must be filled in by authorised personnel and attached to the pump in a visible position.

If Grundfos is requested to service the pump, it must be cleaned before it is returned.

If proper cleaning is not possible, all relevant information about the pumped liquid must be provided.

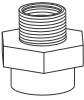
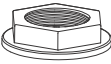
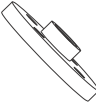
If the above is not fulfilled, Grundfos can refuse to accept the pump for service.

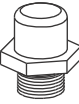
Possible costs of returning the pump are to be paid by the customer.

The safety declaration can be found at the end of these instructions (only in English).

2.3.1 Description of kits

Tank		
Designation	Quantity	
	270 litres	540 litres
Tank	1	1
Cover with gasket	1	2
Screw, M8 x 30	12	24
Washer	12	24
Fastening bracket for level sensor tube	2	2
Screw for fixing the tank	4	4
Expansion anchor for fixing the tank	4	4
Socket seal, DN 40	1	1
Socket seal, DN 100	1	1

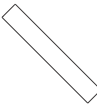
Pump connecting pipes				
Designation		Quantity		
		KP/ CC	SEG	APB
Adapter, Rp 2 - R 1 1/2		1	1	-
Double nipple, R 2 - R 2		1	1	1
Nut, PVC, 2"		1	1	1
O-ring, Ø60, EPDM		-	-	1
Adapter		2	2	2
Stainless-steel clamp		2	2	2
Flexible connecting tube, Ø50		1	1	1
Elbow, 90 °		-	1	1
Flange, Ø50		-	1	-
Screw for flange		-	4	-
Washer for flange		-	8	-

Pump connecting pipes				
Designation		Quantity		
		KP/ CC	SEG	APB
Nut, M16, for flange		-	4	-
O-ring, Ø50, for flange		-	1	-
Unilift KP: Double nipple		1	-	-
Unilift CC: Adapter 1 1/2" - 1 1/4"		1	-	-
Unilift KP: Non-return flap valve		1*		
Thread tape		1	1	1
Vent port				
Socket, PVC, 75 x 63 x 2"		1	1	1
Gasket, EPDM, Ø63		1	1	1
Nut, PVC, 2"		1	1	1
Other components				
Schuko plug		1	-	1
Cable gland, M20		2	1	2
Cable gland, M25		-	1	-

* The kit for pump connecting pipes contains a non-return valve for the Unilift KP pump. The Unilift CC pump is supplied with a non-return valve.

Accessories

We recommend the following accessories.

Outlet pipe		Quantity			
Designation		Unilift		Duolift	
		CC/ KP	APB/ SEG	CC/ KP	APB/ SEG
Non-return valve		-*	1	-*	2
Double nipple		-	1	3	5
Isolating valve		1	1	1	1
Union nut		-	-	2	2
T-piece, PVC, 90 °		-	-	1	1
Elbow, PVC, 90 °		-	-	2	2
Pipe, Ø50, 2 m		-	-	1	1
Thread tape		1	1	1	1

* The Unilift CC pump is supplied with a non-return valve.

3. Installing the product

3.1 General guidelines

The guidelines for correct mechanical installation of lifting station are according to EN 12056-4.

- If the lifting station is installed in a basement with recurrent penetration of groundwater, we recommend that you install a drainage pump in a separate pump sump below floor level. In some countries the drainage pump is mandatory. If there is only a risk of penetrating groundwater, we recommend that you install a float switch outside the lifting station and connect it to a controller which indicates high-level alarm in case of flooding.
- Fix the tank to the floor.
- We recommend that you connect a diaphragm pump to the tank to be able to empty the tank manually.
- All outlet pipes from the lifting station, diaphragm pump and drainage pump must have a backflow loop, i.e. a bend above the local backflow level. Commonly, the highest point of the bend must be above street level. See fig. 2.
- Surface water must not be discharged into the lifting station inside the building. Collect and discharge surface water outside the building.
- Install a non-return valve at the outlet of the lifting station to prevent backflow.
- Fit isolating valves at the inlet and outlet of the lifting station.
- The volume of the outlet pipe above the non-return valve up to the backflow loop must be smaller than the effective tank volume. See pos. 7 in figures 1 and 2.
- In general, a lifting station for black wastewater must be vented above roof level. However, it is permitted to lead the vent pipe into the main building ventilation system. Place special two-way venting valves outside the building.
- If the wastewater is discharged into a collecting line, the collecting line must have a filling ratio of at least $h/d = 0.7$. The collecting line must be at least one nominal diameter bigger after the outlet pipe connection.
- Use 45 ° bends to avoid deposits in the pipes.

3.2 Mechanical installation

3.2.1 Tank inlets

We recommend that you connect the inlet pipes to the top inlets of the tank. This prevents backflow and deposits in the inlet pipe, and enables a high start level of the pump. A high start level increases the effective tank volume and reduces the number of start/stops of the pumps.

3.2.2 Location

Install the lifting station in a properly lit and vented room with 60 cm free space around all parts to be serviced and operated.

The tank must not be exposed to direct sunlight. UV light can affect the properties of its composite material, resulting in a shorter life.

The product is designed for indoor operation. If installed outside, the lifting station must be placed in a closed pit.

The tank can be flooded and may be installed in a basement with risk of penetrating groundwater.

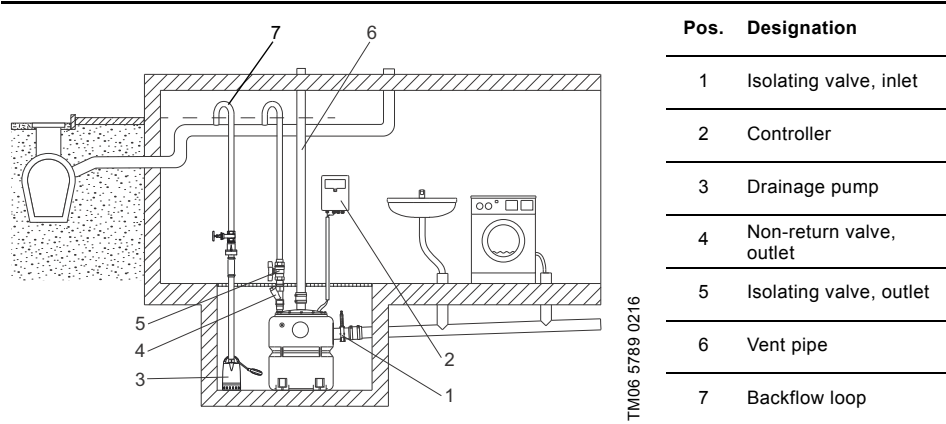


Fig. 1 Installation example with a tank in a sump and with a controller. The diaphragm pump is not shown.

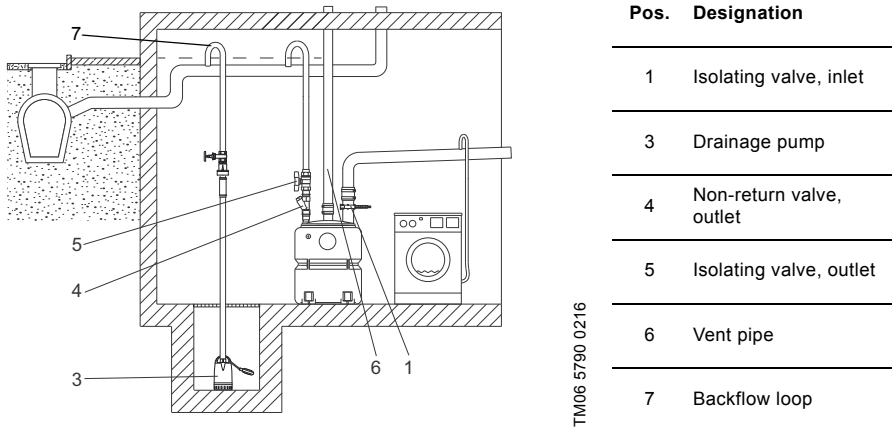


Fig. 2 Installation example with a tank on the basement floor. The diaphragm pump is not shown.

3.3 Tools

- Thread tape
- glue.

Not supplied with the product:

- drilling machine
- drilling bits: $\varnothing 20$, $\varnothing 25$
- hole saw: $\varnothing 40$, $\varnothing 100$
- deburring tool for polyethylene.

3.4 Installation overview

Depending on your product type, you must go through different steps to install your lifting station. You may need to go through additional steps if you have ordered accessories.

Lifting station with pumps with float switches

1. Fix the tank to the floor.
2. Drill holes for the inlet pipes.
3. Connect the inlet pipes to the tank.
4. Fit the vent port and the cable glands.
5. Set the start levels by adjusting the cable length of the float switch.
6. Install the pumps in the tank.
7. Fit the pump connecting pipes.
8. Connect the vent pipe.
9. Connect the outlet pipes.

Lifting station with controller

1. Fix the tank to the floor.
2. Drill holes for the inlet pipes.
3. Connect the inlet pipes to the tank.
4. Fit the vent port and the cable glands.
5. Install the pumps in the tank.
6. Fit the level sensor tube in the tank.
7. Fit the pump connecting pipes.
8. Connect the vent pipe.
9. Connect the outlet pipes.

3.5 Positioning the product

CAUTION

Back injury

- Minor or moderate personal injury
- Get assistance from another person or use lifting equipment to carry the product.

CAUTION

Crushing of feet

- Minor or moderate personal injury
- Wear safety shoes when moving the product.

Position the tank so that all components are easily accessible during operation and maintenance.

3.6 Installing the tank

Fix the tank to the floor to prevent uplift and twist. Use the mounting brackets to fix the tank with screws.

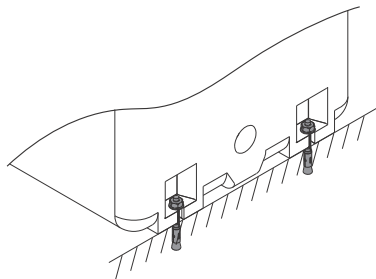


Fig. 3 Fixing the tank to the floor

3.7 Drilling holes for the pipes

CAUTION



Sharp element

- Minor or moderate personal injury
- Wear protective gloves.

The tank has three or four holes, $\varnothing 100$, for the inlet pipes, depending on the size. The tank also has a hole, $\varnothing 40$, for the diaphragm pump. See section [6.1 Product overview](#). These holes must be drilled to be used.

Holes for the outlet pipes, the vent port and the minimum required cable glands are open.

1. Determine which inlets you want to use. See sections [3.2.1 Tank inlets](#) and [7.1 Setting the start levels](#).
2. Drill holes with a hole saw. The diameter of the hole saw must be the same as the diameter of the pipe.

Inlet: $\varnothing 100$

Hole for diaphragm pump: $\varnothing 40$

3. Deburr the holes.

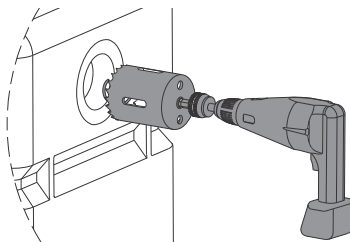


Fig. 4 Drilling holes for pipes

TM06 5780 0116

TM06 5851 0216

3.8 Fitting the cable glands

Choose the appropriate cable gland:

Cable gland for Unilift pumps: M20

Cable gland for SEG pumps: M25

Cable gland for level sensor hose: M20

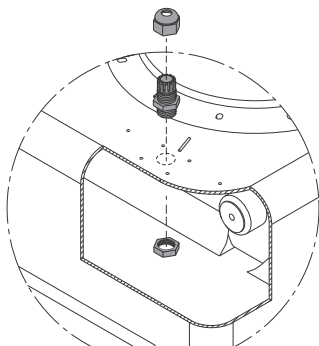


Fig. 5 Fitting the cable glands

3.9 Fitting the vent port

1. Fit the gasket on the socket.
2. Fit the nipple in the hole on top of the tank.
3. Fit the nut inside the tank and tighten it.

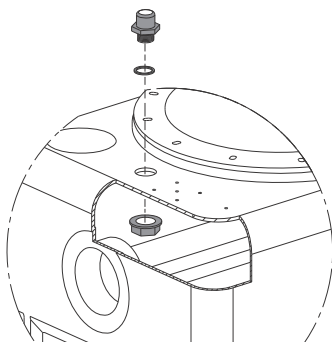


Fig. 6 Fitting the vent port

3.10 Connecting the vent pipe

The vent pipe is not supplied with the product.

You can connect your vent pipe to the vent port in three different ways:

- By means of a PVC vent pipe with an outer diameter of 63 mm and glue.
- By means of a flexible pipe connection with an inner diameter of 75 mm and with clamps. A flexible pipe connection with clamps is available as an accessory.
- By means of a PP double nipple with an inner diameter of 75 mm. The nipple must comply with EN 1451.

3.11 Connecting the inlet pipes

You can connect three or four inlet pipes to the tank, depending on the tank size. First you must drill the holes. See section [3.7 Drilling holes for the pipes](#).

1. Fit the socket seal in the inlet.
2. Connect the isolating valve to the inlet pipe. The isolating valve is an accessory. Product number 96615831.
3. Insert the inlet pipe into the socket seal.

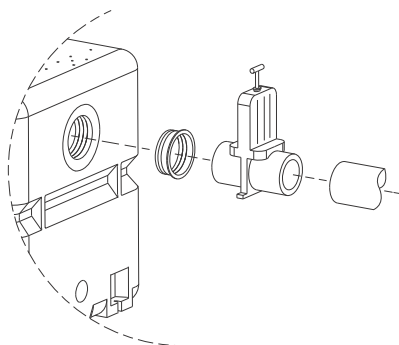


Fig. 7 Connecting an inlet pipe with an isolating valve

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TM06 5924 0316

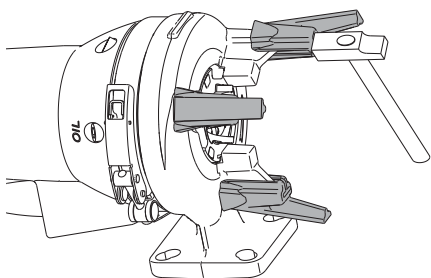
TM06 5859 0216

3.12 Installing the pump

Pumps with float switches

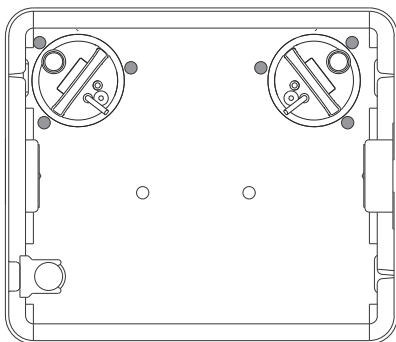
Adjust the cable length of the float switch before you install the pump. See section [7.1.2 Setting the start levels with float switches](#).

1. Remove the plug from the pump cable. You may need to cut the cable.
2. SEG: Fit the foot extensions. See fig. 8.
3. Position the pump in the tank.
Make sure that the float switch, if any, can move freely.
Unilift CC, KP, APB:
Position the pump between the three positioning dots at the bottom of the tank so that it stays in place during operation. See figures 9 and 10.
SEG: The pump weight ensures that the pump stays in place.
4. Lead the pump cable through the cable gland of the tank.
5. Tighten the cable gland.



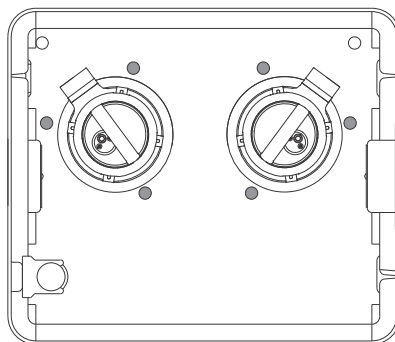
TM02 6571 2116

Fig. 8 Fitting the foot extensions on the SEG pump



TM06 5965 0316

Fig. 9 Correct position of Unilift CC and KP pumps



TM06 5966 0316

Fig. 10 Correct position of Unilift APB pumps

3.13 Installing the pump connecting pipes

3.13.1 Cutting the flexible tube

The connecting pipes are supplied with a flexible tube of $\varnothing 50$. Cut the tube to a length according to the table below.

Pump type	Tube length [mm]
Unilift CC	240
Unilift KP	350
	Not to be cut
Unilift APB	423
SEG	375

3.13.2 Assembling the pump connecting pipes

Use the thread tape in the kit to seal the threaded connections.

1. Assemble the connecting pipe outside the tank, except for the nut. See pos. 1 in fig. 11.
2. Connect the connecting pipe to the pump in the tank.
3. Connect the connecting pipe to the outlet of the tank. Make sure that the pump is in position and that the float switch, if any, can move freely before you tighten the nut.

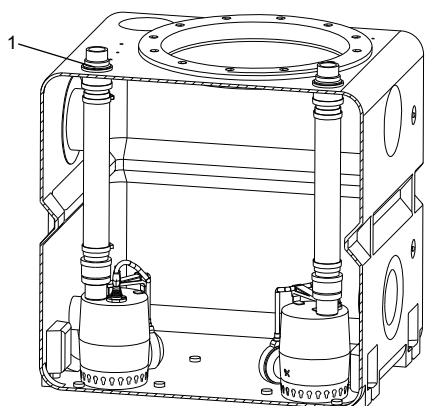


Fig. 11 Example with Unilift KP pumps

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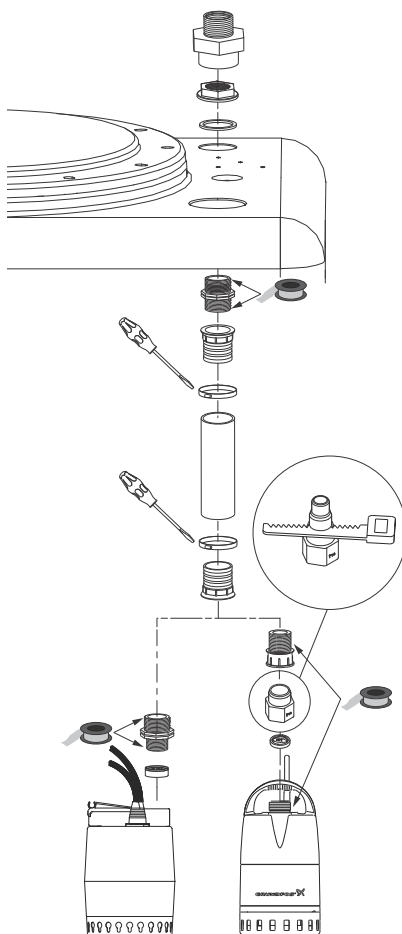


Fig. 12 Unilift CC, KP

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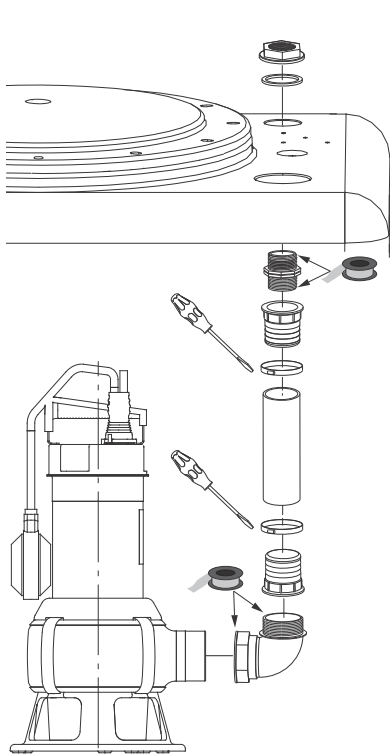


Fig. 13 Unilift AP35B, AP50B

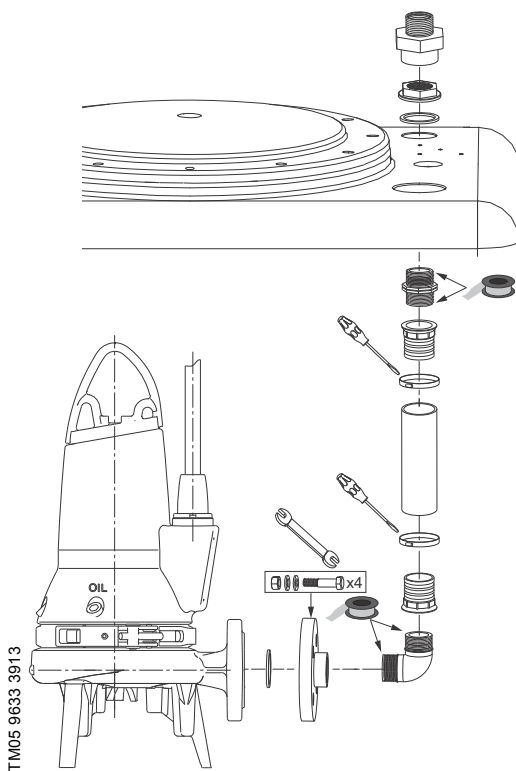


Fig. 14 SEG

TM05 9633 3913

3.14 Installing the level sensor tube

If your pump is not equipped with a float switch, you must install a level sensor tube in the tank.

1. Fix the level sensor tube to the brackets inside the tank.
The locking ring of the tube must be under the top bracket to prevent the tube from rising with the water level.
2. Lead the level sensor hose through the gland.
3. Tighten the gland.

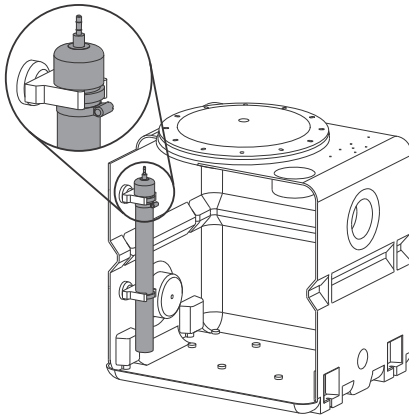


Fig. 15 Correct positioning of the level sensor tube

3.15 Installing an alarm float switch

If you have an alarm device, you can install an additional float switch inside the tank and connect it to the alarm device in order to trigger a high-level alarm.

If you have an LC 220 or LC 221 controller, you do not need an additional alarm float switch as high-level alarm is triggered by the level sensor.

1. Fix the cable of the float switch to the pump connecting pipe with a cable tie.
 - Make sure that there is a free cable length of minimum 70 mm.
 - Make sure that the float switch can move freely from its lowest position to its highest position.
2. Tighten the cable tie.
3. Drill a hole, Ø20, for the cable gland.
4. Fit the cable gland. See fig. 5.
5. Lead the cable of the float switch through the cable gland.
6. Tighten the cable gland.

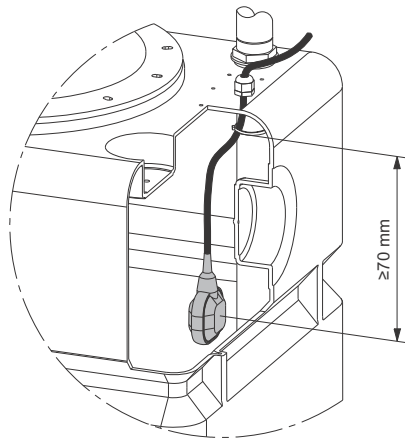


Fig. 16 Alarm float switch

3.16 Installing the outlet pipes

The outlet pipes described in this section are accessories.

3.16.1 Cutting the pipe

This section applies to Duolift. The outlet pipe kit includes a 2 metres' pipe Ø50. Cut the pipe into two pieces of according to the table below.

Model	Pipe length [mm]
Duolift CC, KP, SEG, 270 L	2 x 180
Duolift CC, KP, SEG, 540 L	2 x 430
Duolift APB, 270 L	2 x 163
Duolift APB, 540 L	2 x 413

3.16.2 Assembling the outlet pipes

See the drawings below.

Tips for Duolift

1. Assemble the outlet pipes before connecting anything to the tank.
2. Connect the outlet pipes to the tank by tightening the two union nuts on the tank outlets. See pos. 1 in figures 17 and 19.

TM06 5728 5312

TM06 5899 0216

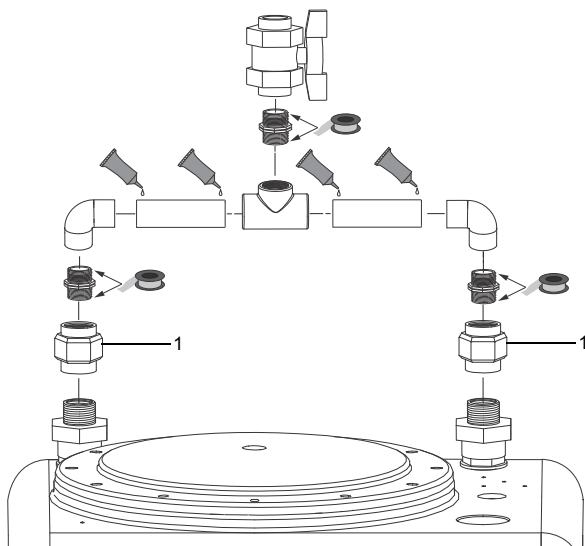


Fig. 17 Duolift CC, KP*

TM06 6220 3913

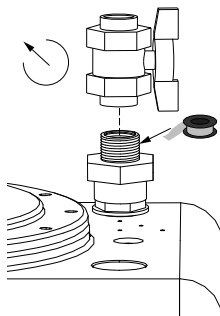


Fig. 18 Unilift CC, KP*

TM06 6512 1816

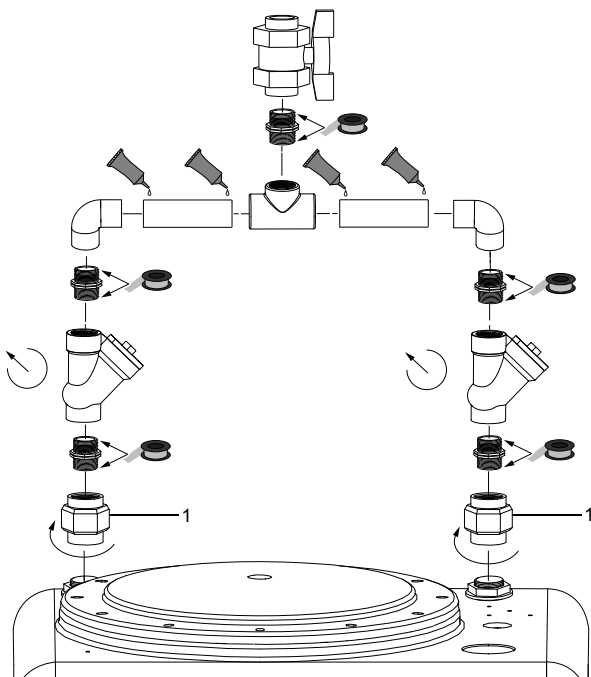


Fig. 19 Duolift APB, SEG**

TM05 9635 3913

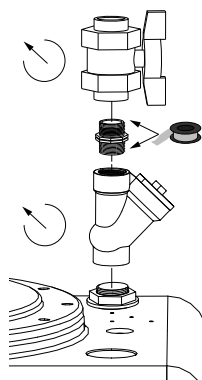


Fig. 20 Unilift APB, SEG**

TM05 9634 3913

* The Unilift CC pump is supplied with a non-return valve.

** The adapter Rp 2 - R 1 1/2 for Duolift SEG is not shown on the drawing.

3.17 Installing a diaphragm pump

The diaphragm pump is optional.

1. Drill the outlet for the diaphragm pump. See section [3.7 Drilling holes for the pipes](#).
2. Fit the DN 40 socket seal in the opening for the diaphragm pump at the bottom of the tank. See figures [21](#) and [22](#).
3. Insert the pipe from the diaphragm pump into the outlet.
4. Secure the diaphragm pump and connect it to the building outlet pipe according to the installation and operating instructions of the pump.

3.18 Installing the controller

Mount the controller on a wall with screws through the mounting holes of the back plate.

The cable entries must point downwards, and the user interface must be easily accessible.

See the installation and operating instructions of the controller.

3.19 Electrical connection

DANGER

Electric shock



Death or serious personal injury

- Before starting any work on the product, make sure that the power supply has been switched off and that it cannot be accidentally switched on.

DANGER

Electric shock



Death or serious personal injury

- The protective earth (PE) of the power outlet must be connected to the protective earth of the product. The plug must have the same PE connection system as the power outlet.

DANGER

Electric shock



Death or serious personal injury

- The installation must be fitted with a residual current device (RCD) with a tripping current less than 30 mA.

DANGER

Electric shock



Death or serious personal injury

- The product must be connected to an external main switch with a minimum contact gap of 3 mm (0.12 inch) in all poles.

3.19.1 Connecting the pump

Connecting the pump to a power socket

1. Lead the pump cable through the cable gland.
2. Fit the power plug on the cable. A Schuko plug is supplied with the product.
3. Insert the plug into a power socket.

Connecting the pump to a controller

1. Lead the pump cable through the cable gland.
2. Connect the pump cable to the controller. See the wiring diagram supplied with the controller.

3.19.2 Connecting the level sensor hose to the controller

Connect the level sensor hose to the pneumatic entry in the bottom of the controller. The hose is connected properly when it is pushed all the way in and locked in position.

3.19.3 Connecting an alarm float switch to the controller

1. Lead the cable of the float switch through one of the M20 cable glands in the bottom of the controller.
2. Connect the cable to the terminal for high-level alarm. See the installation and operating instructions of the controller.

3.20 Setting the controller before startup

LC 220

Select settings via the DIP switches. See the installation and operating instructions of the controller.

LC 221

You must set the start level and rated current of the motor to be able to start the product. All other settings are preset.

1. Close the isolating valves.
2. Set the selector switches on the controller to "OFF".
3. Switch on the controller.
4. When the display shows "L_01", press [OK] to enter the start level menu.
5. Select the start level using the left and right arrows and press [OK]. See section [7.1.1 Setting the start levels with a controller](#).
6. Go to the rated current menu "L_02" using the left and right arrows and select the rated motor current [A] according to the nameplate of the pump motor.

You can change some factory settings. See the installation and operating instructions of the controller.

LC 220



<http://net.grundfos.com/qr/i/98503252>

QR98503252

LC 221



<http://net.grundfos.com/qr/i/98503253>

QR98503253

4. Starting up the product

4.1 Starting up a product with a controller

Make sure that the controller is set correctly. See section [3.20 Setting the controller before startup](#).

1. Make sure that all pipes and connections are tight.
2. Switch on the controller.
3. Set the selector switch to "AUTO".
4. Open the isolating valves.
5. Flush a toilet or open a tap connected to one of the inlets.
6. Monitor the increasing liquid level in the tank up to the start level and check that the pumps start and stop at least twice.

4.2 Starting up a product without a controller

1. Make sure that all pipes and connections are tight.
2. Open the isolating valves.
3. Switch on the power supply to the pump.
4. Flush a toilet or open a tap connected to one of the inlets.
5. Monitor the increasing liquid level in the tank up to the start level and check that the pumps start and stop at least twice.

5. Storing the product

CAUTION

Back injury



- Minor or moderate personal injury
- Get assistance from another person or use lifting equipment to carry the product.

CAUTION

Crushing of feet



- Minor or moderate personal injury
- Wear safety shoes when moving the product.

The tank must not be exposed to direct sunlight. UV light can affect the properties of its composite material, resulting in a shorter life.

For information on how to store the pumps and the controller, see the installation and operating instructions of the product.

6. Product introduction

6.1 Product overview

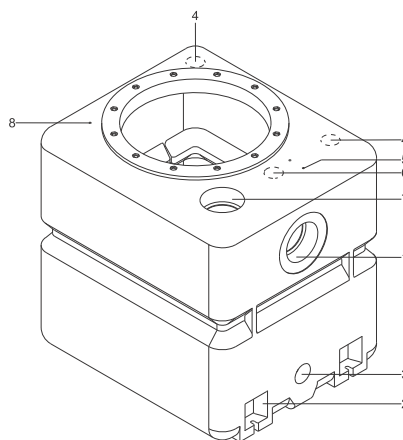


Fig. 21 Unolift 270 / Duolift 270

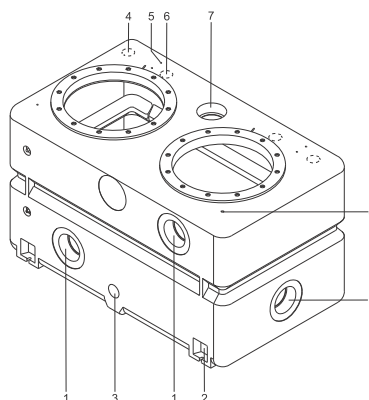


Fig. 22 Duolift 540

Pos.	Description
1	Horizontal main inlets, DN 100
2	Mounting brackets
3	Connection, DN 40, for optional diaphragm pump
4	Pump outlet
5	Holes for cable glands
6	Outlet for vent pipe, Ø60
7	Vertical main inlet, DN 100
8	Hole for level sensor hose

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6.2 Applications

The lifting station is designed for pumping grey and black wastewater from family houses and commercial buildings where the liquid cannot be led directly to the sewer due to lack of a natural downward slope.

6.3 Pumped liquids

Unilift CC, Unilift KP, Unilift APB

Grey wastewater, that is domestic wastewater without toilet discharge.

Unilift AP50B may be used for black wastewater, that is domestic wastewater with toilet discharge, in countries allowing outlet pipes of DN 50 for black wastewater without using a grinder pump.

SEG

Grey and black wastewater, that is domestic wastewater with toilet discharge.

The SEG pump has a grinder system.

Surface water must not be discharged into a lifting station inside the building.

6.4 Automatic operation

The lifting station is designed to operate automatically according to the liquid level in the tank. You can set the lifting station to automatic operation in two ways. Equip the lifting station either with float switches, fitted directly on the pumps, or with a level sensor tube and a controller.

6.5 Identification

6.5.1 Type key

UNILIFT 270. 40. CC			
Model	Unilift: one pump Duolift: two pumps		
Tank volume [litres]	270 540		
Outlet	DN 40		
Pump	CC KP APB, APB SEG		

6.5.2 Nameplate

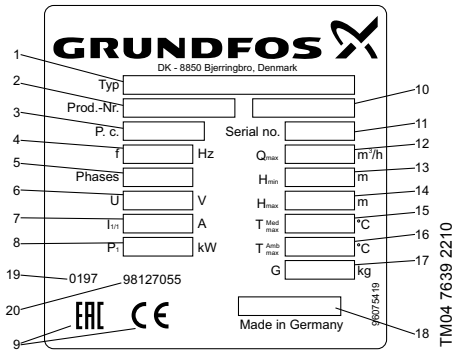


Fig. 23 Nameplate, lifting station

Pos.	Description
1	Type designation
2	Product number
3	Production code (year and week)
4-8	-
9	CE mark
10	-
11	Serial number
12-14	-
15	Maximum liquid temperature [°C]
16	Maximum ambient temperature [°C]
17	Weight [kg]
18	-
19	Reference number for the declaration of performance
20	Notified body

6.6 Diaphragm pump

We recommend that you connect a diaphragm pump to the tank. It is a positive displacement pump which can be operated manually. It can be used to drain the tank before service or during power cuts. The tank has a DN 40 outlet for the pump.

6.7 Level sensor tube

The level sensor is an accessory, which works together with a controller LC 220 or LC 221. The level sensor tube is connected to a piezoresistive level sensor in the controller via the level sensor hose. The rising liquid level compresses the air inside the level sensor tube and hose, and the piezoresistive sensor transforms the changing pressure into an analog signal. The controller uses the analog signal to start and stop the pump and to indicate high-level alarm.

7. Operating the product

The lifting station is designed to operate automatically according to the liquid level in the tank. With a controller, you can easily change settings, monitor the system or perform fault finding.

You can download the installation and operating instructions of the controller with the following links and QR codes.

LC 220



QR98503252

<http://net.grundfos.com/qr/i/98503252>

LC 221



QR98503253

<http://net.grundfos.com/qr/i/98503253>

7.1 Setting the start levels

The optimum start level is closely related to the chosen inlet height and pipe configuration. The start level must be as high as possible to increase the effective tank volume and reduce the number of start/stops of the pumps. A too high start level, however, may increase the backflow in the inlet pipe and thus increase the risk of deposits in the inlet pipe.

The figures below show the recommended start levels in relation to the inlet height and pipe configuration.

You can select all the values with LC 221. With LC 220, you can select the lowest or highest value.

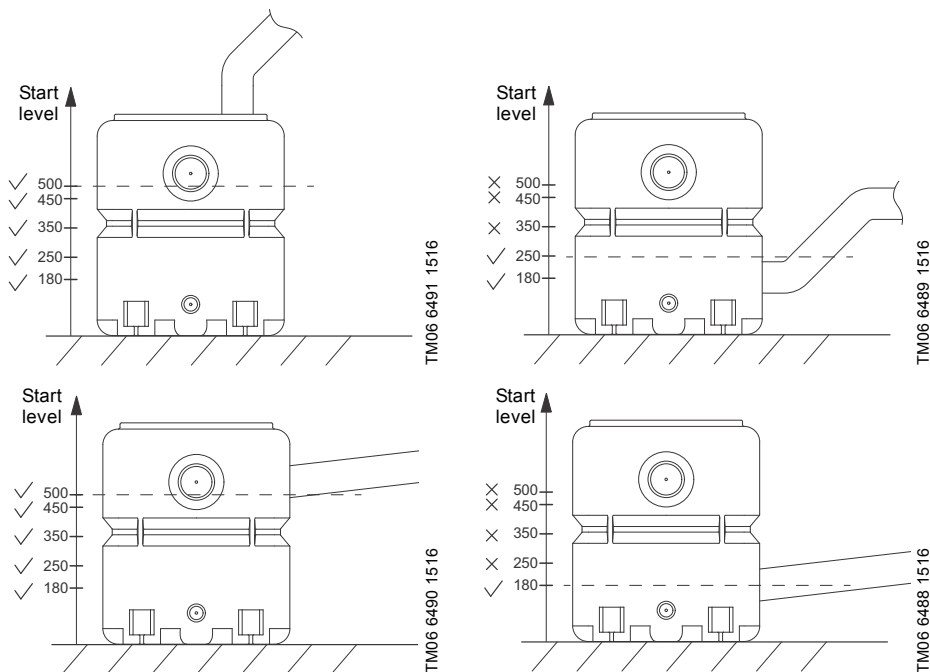


Fig. 24 Recommended start levels in relation to the inlet height and pipe configuration

7.1.1 Setting the start levels with a controller

LC 220

Select a start level, either "LOW" or "HIGH", via the DIP switches. The stop and alarm levels are set automatically according to the selected start level.

	LOW	HIGH
Start level [mm]	180	500
Stop level [mm]	100	100
Alarm level [mm]	350	600

See the installation and operating instructions of the controller for more information about the DIP switches.

You can change the values corresponding to "LOW" and "HIGH" via PC Tool, which is available as an accessory. Product number: 96705378.

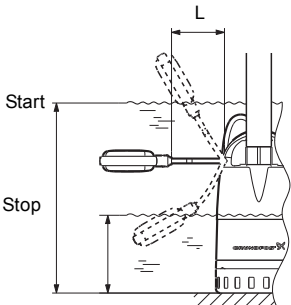
LC 221

Go go [Setup], [L_01] and select one of the following start levels. The stop and alarm levels are set automatically according to the selected start level.

	1	2	3	4	5
Start level [mm]	180	250	350	450	500
Stop level [mm]	100	100	100	100	100
Alarm level [mm]	350	400	550	600	600

7.1.2 Setting the start levels with float switches

Set the start and stop levels by adjusting the free cable length (L).



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Fig. 25 Example of start and stop levels for Unilift CC

We recommend start and stop levels corresponding to a cable length between the minimum and maximum values indicated below.

Unilift KP	Cable length (L)			
	70 mm (minimum)		150 mm (maximum)	
	Start	Stop	Start	Stop
Unilift KP 150	290	140	335	100
Unilift KP 250				
Unilift KP 350	300	150	345	110

Unilift CC	Cable length (L)			
	100 mm (minimum)		200 mm (maximum)	
	Start	Stop	Start	Stop
Unilift CC 5	350	115	400	55
Unilift CC 7				
Unilift CC 9	350	150	435	90

Unilift APB	Cable length (L)			
	100 mm (minimum)		400 mm (maximum)	
	Start	Stop	Start	Stop
Unilift AP35/50B	430	320	460	140

7.1.3 Setting the start levels with a lever arm

	Lever arm position	Start [mm]	Stop [mm]
Unilift KP 150	High*	250	50
Unilift KP 250			
Unilift KP 350			
Unilift CC 5	-	211	89
Unilift CC 7	-		
Unilift CC 9	-	247	125
	Low		
Unilift AP35/50B	Middle	330	210
	High	380	260

* Unilift KP: Use only the high position. See fig. 26.

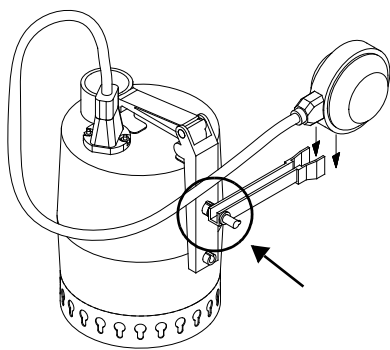


Fig. 26 Lever arm on Unilift KP

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8. Starting up the product after standstill

If the lifting station has been stopped for a long time, for example in a holiday house during winter, you must remove any hard deposits from the bottom of the tank.

If you hear an unusual noise when the pumps start, the pumps may be blocked. Close the isolating valve on the inlet side and remove the blockage before you start the system again. You may need to dismantle the pump. See section 12. [Fault finding the product](#).

9. Servicing the product

WARNING

Biological hazard



Death or serious personal injury

- Wear protective gloves, close the isolating valves and drain the tank before you start any work on the tank.

According to EN 12056, lifting stations in family houses must be maintained at least once a year, in blocks of flats twice a year and in commercial buildings four times a year.

We recommend that you enter a maintenance contract with Grundfos to ensure a consistent operational safety of the system.

Regular checks of the system must include the following:

- Check for deposits at the bottom of the tank.
- Check for leakage.
- Check that the float switch, if any, can move freely.
- Check that the pump is in correct position.

For pump maintenance, see the installation and operating instructions of the pump. It is easy to remove the pumps from the tank thanks to the flexible connecting pipes and clamps.

10. Taking the product out of operation

WARNING

Biological hazard



Death or serious personal injury

- Wear protective gloves, close the isolating valves and drain the tank before you start any work on the tank.

CAUTION

Sharp element



Minor or moderate personal injury

- Wear protective gloves when handling the pump. The impeller has sharp edges.

CAUTION

Crushing of feet



Minor or moderate personal injury

- Wear safety shoes when moving the product.

1. Close the isolating valves on the inlet side.
2. Start the pumps manually to empty most of the tank.
3. Close the isolating valve on the outlet side.
4. Empty the tank completely with the diaphragm pump.
5. Switch of the power supply to the pumps or to the controller, if any.

You can now disconnect and dismantle all the pipes, remove the pumps, cables, etc.

11. Technical data

11.1 Operating conditions

Ambient temperature	0-40 °C
Liquid temperature	0-40 °C during normal operation. Maximum 60 °C for 5 minutes.
Maximum system pressure	6 bar

11.2 Material

Tank	PE
Level sensor tube	PVC
Parts for pipe connection	PVC/PP/EPDM

11.3 Electrical data

See the installation and operating instructions of the pump and the controller.

11.4 Dimensions and weights

The weight applies to the tank only.

	Volume [litres]	Weight [kg]
Small tank	270	19.5
Big tank	540	29.5
Tank cover	-	2

12. Fault finding the product

WARNING

Biological hazard



- Death or serious personal injury
- Wear protective gloves, close the isolating valves and drain the tank before you start any work on the tank.

For faults indicated by alarms and related to the controller, see the installation and operating instructions of the controller.

For faults related to the pumps, see the installation and operating instructions of the pumps.

Fault	Cause	Remedy
1. The pump does not start.	a) The float switch is blocked in low position.	1. Make sure that the float switch can move freely. 2. It may be necessary to adjust the cable length. See section 7.1 Setting the start levels .
	b) The pump cable is defective.	Repair or replace the cable.
2. The pump does not stop.	a) The float switch is blocked in high position.	1. Make sure that the float switch can move freely. 2. It may be necessary to adjust the cable length. See section 7.1 Setting the start levels .
3. The pump starts and stops randomly.	a) The float switch is not set properly.	Adjust the cable length of the float switch. See section 7.1 Setting the start levels .
	b) The level sensor tube is not fixed properly in the tank and is moving up as the liquid rises. The sensor receives wrong signals.	Fix the tube to the tank. See fig. 13 .
	c) The float switch cannot move freely because the pump has moved.	Position the pump according to fig. 9 or 10 .
4. The tank is not emptying although the pump is running.	a) There is an airlock in the pump impeller. The pump cannot build up pressure.	Drill a hole of maximum 6 mm in the connecting pipes in the tank. The air is pushed out when the pump starts, and the pump builds up pressure. The pressure loss due to the hole is negligible. Do not drill a hole in the outlet pipe outside the tank.
5. The pump makes an unusual sound.	a) The pump is blocked.	Take the pump out of the tank and remove the blockage. You may need to dismantle the pump. See the installation and operating instructions of the pump.

13. Disposing of the product

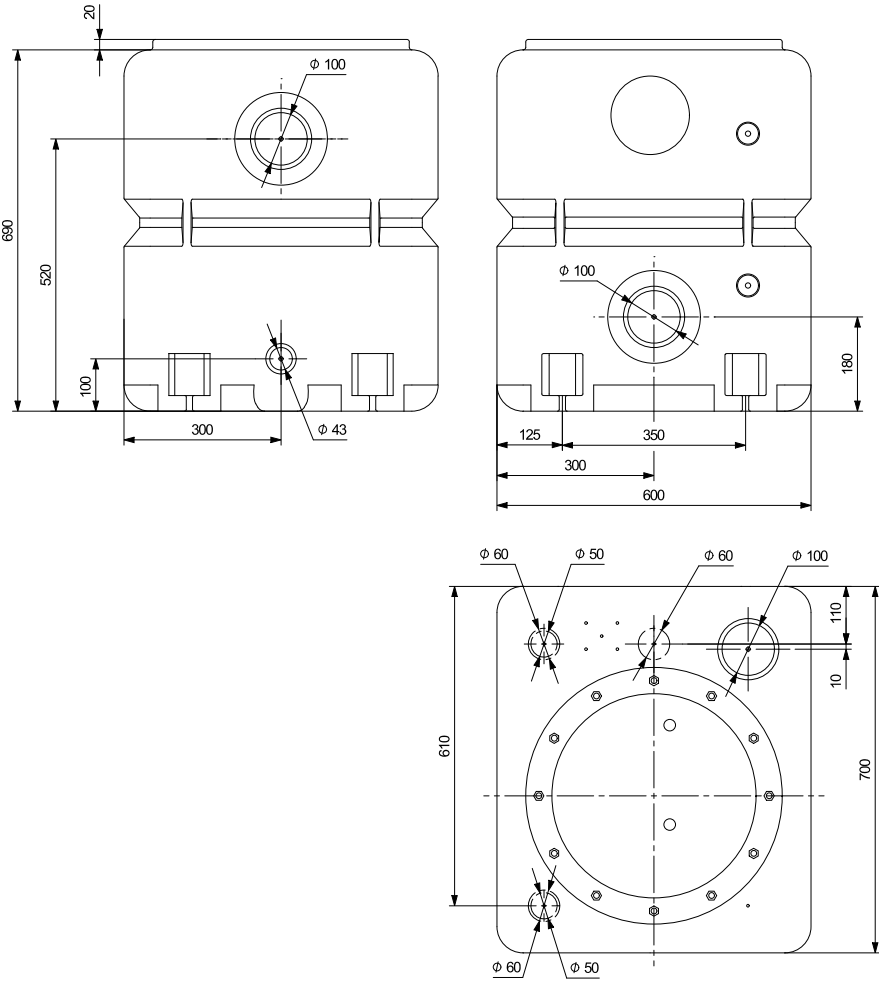
This product or parts of it must be disposed of in an environmentally sound way:

1. Use the public or private waste collection service.
2. If this is not possible, contact the nearest Grundfos company or service workshop.



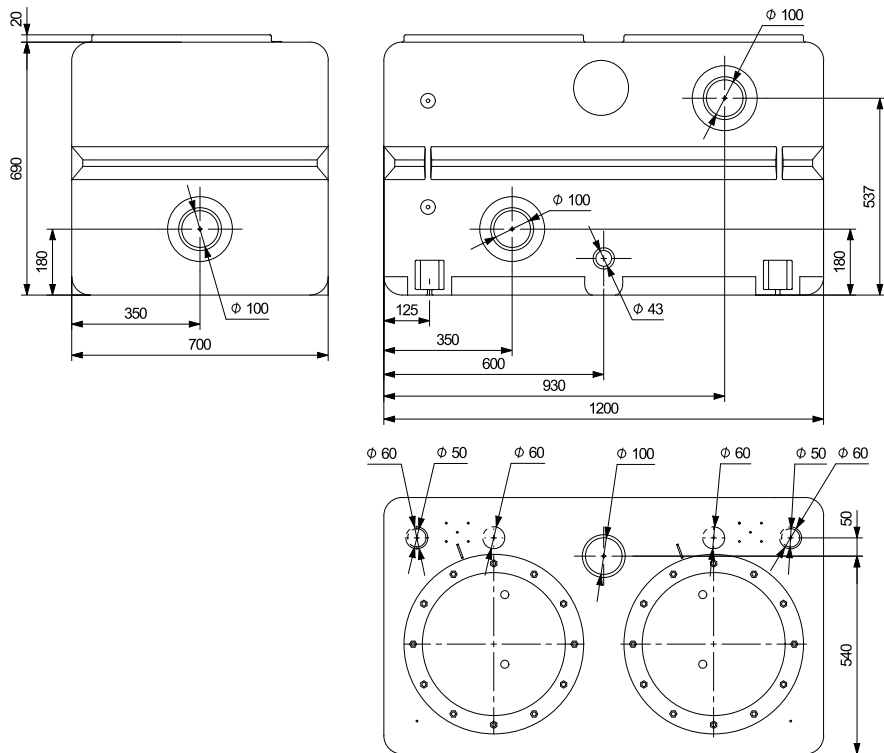
The crossed-out wheellie bin symbol on a product means that it must be disposed of separately from household waste. When a product marked with this symbol reaches its end of life, take it to a collection point designated by the local waste disposal authorities. The separate collection and recycling of such products will help protect the environment and human health.

Appendix



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Fig. 1 Unolift 270, Duolift 270



TM06 1066 1514

Fig. 2 Duolift 540

Argentina

Bombas GRUNDFOS de Argentina S.A.
Ruta Panamericana km. 37.500 Centro
Industrial Garin
1619 Garin Pcia. de B.A.
Phone: +54-3327 414 444
Telefax: +54-3327 45 3190

Australia

GRUNDFOS Pumps Pty. Ltd.
P.O. Box 2040
Regency Park
South Australia 5942
Phone: +61-8-8461-4611
Telefax: +61-8-8340 0155

Austria

GRUNDFOS Pumpen Vertrieb
Ges.m.b.H.
Grundfosstraße 2
A-5082 Grödig/Salzburg
Tel.: +43-6246-883-0
Telefax: +43-6246-883-30

Belgium

N.V. GRUNDFOS Bellux S.A.
Boomsesteenweg 81-83
B-2630 Aartselaar
Tél.: +32-3-870 7300
Télécopie: +32-3-870 7301

Belarus

Представительство ГРУНДФОС в
Минске
220125, Минск
ул. Шафарнянская, 11, оф. 56, БЦ
«Порт»
Тел.: +7 (375 17) 286 39 72/73
Факс: +7 (375 17) 286 39 71
E-mail: minsk@grundfos.com

Bosnia and Herzegovina

GRUNDFOS Sarajevo
Zmaja od Bosne 7-7A,
BH-71000 Sarajevo
Phone: +387 33 592 480
Telefax: +387 33 590 465
www.ba.grundfos.com
e-mail: grundfos@bih.net.ba

Brazil

BOMBAS GRUNDFOS DO BRASIL
Av. Humberto de Alencar Castelo
Branco, 630
CEP 09850 - 300
São Bernardo do Campo - SP
Phone: +55-11 4393 5533
Telefax: +55-11 4343 5015

Bulgaria

Grundfos Bulgaria EOOD
Slatina District
Iztochna Tangenta street no. 100
BG - 1592 Sofia
Tel. +359 2 49 22 200
Fax. +359 2 49 22 201
email: bulgaria@grundfos.bg

Canada

GRUNDFOS Canada Inc.
2941 Brighton Road
Oakville, Ontario
L6H 6C9
Phone: +1-905 829 9533
Telefax: +1-905 829 9512

China

GRUNDFOS Pumps (Shanghai) Co. Ltd.
10F The Hub, No. 33 Suhong Road
Minhang District
Shanghai 201106
PRC
Phone: +86 21 612 252 22
Telefax: +86 21 612 253 33

COLOMBIA

GRUNDFOS Colombia S.A.S.
Km 1.5 vía Siberia-Cota Conj. Potrero
Chico,
Parque Empresarial Arcos de Cota Bod.
1A.
Cota, Cundinamarca
Phone: +57(1)-2913444
Telefax: +57(1)-8764586

Croatia

GRUNDFOS CROATIA d.o.o.
Buzinski prilaz 38, Buzin
HR-10010 Zagreb
Phone: +385 1 6595 400
Telefax: +385 1 6595 499
www.hr.grundfos.com

GRUNDFOS Sales Czechia and Slovakia s.r.o.

Čajkovského 21
779 00 Olomouc
Phone: +420-585-716 111

Denmark

GRUNDFOS DK A/S
Martin Bachs Vej 3
DK-8850 Bjerringbro
Tlf.: +45-87 50 50 50
Telefax: +45-87 50 51 51
E-mail: info_GDK@grundfos.com
www.grundfos.com/DK

Estonia

GRUNDFOS Pumps Eesti OÜ
Peterburi tee 92G
11415 Tallinn
Tel: + 372 606 1690
Fax: + 372 606 1691

Finland

OY GRUNDFOS Pumput AB
Trukkikuja 1
FI-01360 Vantaa
Phone: +358-(0) 207 889 500

France

Pompes GRUNDFOS Distribution S.A.
Parc d'Activités de Chesnes
57, rue de Malacombe
F-38290 St. Quentin Fallavier (Lyon)
Tél.: +33-4 74 82 15 15
Télécopie: +33-4 74 94 10 51

Germany

GRUNDFOS GMBH
Schlüterstr. 33
40699 Erkrath
Tel.: +49-(0) 211 929 69-0
Telefax: +49-(0) 211 929 69-3799
e-mail: infoservice@grundfos.de
Service in Deutschland:
e-mail: kundendienst@grundfos.de

Greece

GRUNDFOS Hellas A.E.B.E.
20th km. Athinon-Markopoulou Av.
P.O. Box 71
GR-19002 Peania
Phone: +0030-210-66 83 400
Telefax: +0030-210-66 46 273

Hong Kong

GRUNDFOS Pumps (Hong Kong) Ltd.
Unit 1, Ground floor
Siu Wai Industrial Centre
29-33 Wing Hong Street &
68 King Lam Street, Cheung Sha Wan
Kowloon
Phone: +852-27861706 / 27861741
Telefax: +852-27858664

Hungary

GRUNDFOS Hungária Kft.
Tópark u. 8
H-2045 Törökbálint,
Phone: +36-23 511 110
Telefax: +36-23 511 111

India

GRUNDFOS Pumps India Private
Limited
118 Old Mahabalipuram Road
Thoraipakkam
Chennai 600 096
Phone: +91-44 2496 6800

Indonesia

PT. GRUNDFOS POMPA
Graha Intirub Lt. 2 & 3
Jln. Cililitan Besar No.454. Makasar,
Jakarta Timur
ID-Jakarta 13650
Phone: +62 21-469-51900
Telefax: +62 21-460 6910 / 460 6901

Ireland

GRUNDFOS (Ireland) Ltd.
Unit A, Merrywell Business Park
Ballymount Road Lower
Dublin 12
Phone: +353-1-4089 800
Telefax: +353-1-4089 830

Italy

GRUNDFOS Pompe Italia S.r.l.
Via Gran Sasso 4
I-20060 Truccazzano (Milano)
Tel.: +39-02-95838112
Telefax: +39-02-95309290 / 95838461

Japan

GRUNDFOS Pumps K.K.
1-2-3, Shin-Miyakoda, Kita-ku,
Hamamatsu
431-2103 Japan
Phone: +81 53 428 4760
Telefax: +81 53 428 5005

Korea

GRUNDFOS Pumps Korea Ltd.
6th Floor, Aju Building 679-5
Yeoksam-dong, Kangnam-ku, 135-916
Seoul, Korea
Phone: +82-2-5317 600
Telefax: +82-2-5633 725

Latvia

SIA GRUNDFOS Pumps Latvia
Deglava biznesa centrs
Augusta Deglava ielā 60, LV-1035, Rīga,
Tālr.: + 371 714 9640, 7 149 641
Fakss: + 371 914 9646

Lithuania

GRUNDFOS Pumps UAB
Smolensko g. 6
LT-03201 Vilnius
Tel: + 370 52 395 430
Fax: + 370 52 395 431

Malaysia

GRUNDFOS Pumps Sdn. Bhd.
7 Jalan Peguam U1/25
Glenmarie Industrial Park
40150 Shah Alam
Selangor
Phone: +60-3-5569 2922
Telefax: +60-3-5569 2866

Mexico

Bombas GRUNDFOS de México S.A. de C.V.
Boulevard TLC No. 15
Parque Industrial Stiva Aeropuerto
Apodaca, N.L. 66600
Phone: +52-81-8144 4000
Telefax: +52-81-8144 4010

Netherlands

GRUNDFOS Netherlands
Veluwezoom 35
1326 AE Almere
Postbus 22015
1302 CA ALMERE
Tel.: +31-88-478 6336
Telefax: +31-88-478 6332
E-mail: info_gnl@grundfos.com

New Zealand

GRUNDFOS Pumps NZ Ltd.
17 Beatrice Tinsley Crescent
North Harbour Industrial Estate
Albany, Auckland
Phone: +64-9-415 3240
Telefax: +64-9-415 3250

Norway

GRUNDFOS Pumper A/S
Strømsveien 344
Postboks 235, Leirdal
N-1011 Oslo
Tlf.: +47-22 90 47 00
Telefax: +47-22 32 21 50

Poland

GRUNDFOS Pompy Sp. z o.o.
ul. Klonowa 23
Baranowo k. Poznań
PL-62-081 Przeźmierowo
Tel: (+48-61) 650 13 00
Fax: (+48-61) 650 13 50

Portugal

Bombas GRUNDFOS Portugal, S.A.
Rua Calvet de Magalhães, 241
Apartado 1079
P-2770-153 Paço de Arcos
Tel.: +351-21-440 76 00
Telefax: +351-21-440 76 90

Romania

GRUNDFOS Pompe România SRL
Bd. Biruintei, nr 103
Pantelimon county Ilfov
Phone: +40 21 200 4100
Telefax: +40 21 200 4101
E-mail: romania@grundfos.ro

Russia

ООО Грундфос Россия
ул. Школьная, 39-41
Москва, RU-109544, Russia
Тел. (+7) 495 564-88-00 (495) 737-30-00
Факс (+7) 495 564 8811
E-mail grundfos.moscow@grundfos.com

Serbia

Grundfos Srbija d.o.o.
Omladinskih brigada 90b
11070 Novi Beograd
Phone: +381 11 2258 740
Telefax: +381 11 2281 769
www.rs.grundfos.com

Singapore

GRUNDFOS (Singapore) Pte. Ltd.
25 Jalan Tukang
Singapore 619264
Phone: +65-6681 9688
Telefax: +65-6681 9689

Slovakia

GRUNDFOS s.r.o.
Prievozská 4D
821 09 BRATISLAVA
Phona: +421 2 5020 1426
sk.grundfos.com

Slovenia

GRUNDFOS LJUBLJANA, d.o.o.
Leskoškova 9e, 1122 Ljubljana
Phone: +386 (0) 1 568 06 10
Telefax: +386 (0) 1 568 06 19
E-mail: tehnika-si@grundfos.com

South Africa

GRUNDFOS (PTY) LTD
Corner Mountjoy and George Allen
Roads
Wilbart Ext. 2
Bedfordview 2008
Phone: (+27) 11 579 4800
Fax: (+27) 11 455 6066
E-mail: lsmart@grundfos.com

Spain

Bombas GRUNDFOS España S.A.
Camino de la Fuenteclilla, s/n
E-28110 Algete (Madrid)
Tel.: +34-91-848 8800
Telefax: +34-91-628 0465

Sweden

GRUNDFOS AB
Box 333 (Lunnagårdsgatan 6)
431 24 Mölndal
Tel.: +46 31 332 23 000
Telefax: +46 31 331 94 60

Switzerland

GRUNDFOS Pumpen AG
Bruggacherstrasse 10
CH-8117 Fällanden/ZH
Tel.: +41-44-806 8111
Telefax: +41-44-806 8115

Taiwan

GRUNDFOS Pumps (Taiwan) Ltd.
7 Floor, 219 Min-Chuan Road
Taichung, Taiwan, R.O.C.
Phone: +886-4-2305 0868
Telefax: +886-4-2305 0878

Thailand

GRUNDFOS (Thailand) Ltd.
92 Chaloen Phrakiat Rama 9 Road,
Dokmai, Pravej, Bangkok 10250
Phone: +66-2-725 8999
Telefax: +66-2-725 8998

Turkey

GRUNDFOS POMPA San. ve Tic. Ltd. Sti.
Gebze Organize Sanayi Bölgesi
İhsan dede Caddesi,
2. yol 200. Sokak No. 204
41490 Gebze/ Kocaeli
Phone: +90 - 262-679 7979
Telefax: +90 - 262-679 7905
E-mail: satis@grundfos.com

Ukraine

Бізнес Центр Європа
Столичне шосе, 103
м. Київ, 03131, Україна
Телефон: (+38 044) 237 04 00
Факс.: (+38 044) 237 04 01
E-mail: ukraine@grundfos.com

United Arab Emirates

GRUNDFOS Gulf Distribution
P.O. Box 16768
Jebel Ali Free Zone
Dubai
Phone: +971 4 8815 166
Telefax: +971 4 8815 136

United Kingdom

GRUNDFOS Pumps Ltd.
Grovebury Road
Leighton Buzzard/Beds. LU7 4TL
Phone: +44-1525-850000
Telefax: +44-1525-850011

U.S.A.

GRUNDFOS Pumps Corporation
9300 Loiret Blvd.
Lenexa, Kansas 66219
Phone: +1-913-227-3400
Telefax: +1-913-227-3500

Uzbekistan

Grundfos Tashkent, Uzbekistan The
Representative Office of Grundfos
Kazakhstan in Uzbekistan
38a, Oybek street, Tashkent
Телефон: (+998) 71 150 3290 / 71 150 3291
Факс: (+998) 71 150 3292

Addresses Revised 14.03.2018

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